

Verona through Rome

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1 Introduction

We present $Roma^M$, a more complete, fully formalized version of $Rome^M$.

2 Model

The configuration and semantics remain unchanged from the report. They can be found in the appendix.

3 Invariants

We present the invariants again both informally and formally, as there have been invariants added:

L1 Objects are globally unique.

L2 Every frame in the stack contains an open region.

H1 Every region contains its bridge object.

H2 There is at most one reference to a region.

H3 References from within a region to other objects remain within that region.

S1 Regions on the stack are unique.

S2 Every reference from a frame into another frame must travel to a frame no later than the origin frame.

S3 Every reference from a frame into a region must be the region of a frame no later than the origin frame.

HS1 Every reference to an object must be to an object that exists in the configuration.

HS2 Every reference to a region must be to a region that exists in the configuration.

Definition 3.1 (Name). A configuration σ satisfies P if

$$\forall x. \quad |\{ y \mid y \in X(\sigma), y = x \}| \leq 1.$$

Definition 3.2 (Name).

$$\forall x \in X(\sigma). \exists y. P(x, y) \wedge Q(y).$$

4 Reachability

Definition 4.1 (Step relation).

$$\frac{P(\sigma, a) \quad Q(a, f, b)}{\sigma \vdash a \xrightarrow{f} b}$$

Definition 4.2 (Reachable).

$$\text{Reachable}(\sigma, i, b) \iff \exists a. \text{Root}(\sigma, i, a) \wedge \sigma \vdash a \xrightarrow{*} b.$$

Definition 4.3 (Two-branch step relation).

$\frac{\text{STEP-OBJECT} \quad P(\sigma, a) = c \quad c.f = b}{\sigma \vdash a \xrightarrow{r} b}$	$\frac{\text{STEP-REGION} \quad Q(\sigma, a) = c \quad \text{guard}(c) \quad b = \text{target}(c)}{\sigma \vdash a \xrightarrow{r} b}$
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Definition 4.4 (Inductively-closed reachability).

$$\begin{array}{c} \text{BASE-A} \\ \frac{x \in X(\sigma) \quad P(x) = a}{\text{Reach}(\sigma, i, a)} \end{array} \quad \begin{array}{c} \text{BASE-B} \\ \frac{x \in X(\sigma) \quad Q(x) = a}{\text{Reach}(\sigma, i, a)} \end{array} \quad \begin{array}{c} \text{STEP} \\ \frac{\sigma \vdash a \xrightarrow{f} b \quad \text{Reach}(\sigma, i, a)}{\text{Reach}(\sigma, i, b)} \end{array}$$

THEOREM 4.5 (NAME). *Statement.*

THEOREM 4.6 (NAME). *Let σ be such that $P(\sigma)$. Then*

$$Q(\sigma) \implies R(\sigma).$$

5 Related Work

6 Conclusion

A Configuration

$$\begin{aligned} \text{RegionId}, \text{ObjectId}, \text{Index} &\subseteq \text{Int} \\ \text{VarName}, \text{FieldName} &\subseteq \text{String} \end{aligned}$$

$$\text{RuntimeConfiguration} \triangleq \text{Heap} \times \text{Stack}$$

$$\begin{aligned} \text{Heap} &\triangleq \text{RegionId} \rightarrow \text{Region} \\ \text{Region} &\triangleq \text{ObjectId} \times \text{ObjMap} \times \text{Status} \\ \text{Status} &\triangleq \text{Open} \mid \text{Closed} \end{aligned}$$

$$\begin{aligned} \text{Stack} &\triangleq \text{List Frame} \\ \text{Frame} &\triangleq \text{RegionId} \times \text{VarName} \times \text{ObjMap} \times \text{VarMap} \times \text{Index} \end{aligned}$$

$$\text{ObjMap} \triangleq \text{ObjectId} \rightarrow \text{FieldMap}$$

$$\begin{aligned} \text{Object} &\triangleq \text{FieldMap} \\ \text{FieldMap} &\triangleq \text{FieldName} \rightarrow \text{Reference} \\ \text{VarMap} &\triangleq \text{VarName} \rightarrow \text{Reference} \end{aligned}$$

$$\text{Reference} \triangleq \text{RId}(\text{RegionId}) \mid \text{Old}(\text{ObjectId})$$

B Operational Semantics

VAR-ASGN-STACK

$$\frac{\begin{array}{c} F = (\text{rid}, \text{bvar}, \text{omap}, \text{vmap}, \text{fid}) \\ x \neq \text{bvar} \quad \text{resolveV}(x, \langle (S :: F), H \rangle) = \epsilon \quad \text{resolveFA}(y.f, \langle (S :: F), H \rangle) = \text{Old}(\text{oid}) \end{array}}{\langle (S :: F), H \rangle \xrightarrow{x := y.f} \langle S :: (\text{rid}, \text{bvar}, \text{omap}, \text{vmap}[x \mapsto \text{Old}(\text{oid})], \text{fid}), H \rangle}$$

VAR-ASGN-BRIDGE

$$\frac{\begin{array}{c} F = (\text{rid}, x, _, _, _) \quad \text{resolveFA}(y.f, \langle (S :: F), H \rangle) = \text{Old}(\text{oid}) \\ \text{Old}(\text{oid}).\text{loc}(\langle (S :: F), H \rangle) = \text{Rgn}(\text{rid}) \quad H(\text{rid}) = (\text{boid}, \text{omap}, \text{status}) \end{array}}{\langle (S :: F), H \rangle \xrightarrow{x := y.f} \langle (S :: F), H[\text{rid} \mapsto (\text{oid}, \text{omap}, \text{status})] \rangle}$$

FIELD-ASGN-STACK

$$\begin{array}{c}
F = (rid, bvar, omap, vmap, fid) \\
\text{resolveV}(x, \langle (S :: F), H \rangle) = \text{Old}(oid) \quad \text{Old}(oid).loc(\langle (S :: F), H \rangle) = \text{Stk}(fid) \\
\text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(oid') \quad omap(oid) = obj \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: (rid, bvar, omap[oid \mapsto obj[f \mapsto \text{Old}(oid')]], vmap, fid)), H \rangle
\end{array}$$

FIELD-ASGN-REGION

$$\begin{array}{c}
F = (rid, _, _, _) \quad \text{resolveV}(x, \langle (S :: F), H \rangle) = \text{Old}(oid) \\
\text{Old}(oid).loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \quad \text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(oid') \\
\text{Old}(oid').loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \quad H(rid) = (boid, omap, Open) \quad omap(oid) = obj \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F), H[rid \mapsto (boid, omap[oid \mapsto obj[f \mapsto \text{Old}(oid')]], Open)] \rangle
\end{array}$$

SWAP-STACK

$$\begin{array}{c}
F = (rid, bvar, omap, vmap, fid) \\
x \neq bvar \quad vmap(x) = xref \quad \text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \\
\text{resolveFA}(y.f, \langle (S :: F), H \rangle) = yfref \quad \text{Old}(yoid).loc(\langle (S :: F), H \rangle) = \text{Stk}(fid) \\
omap(yoid) = obj \quad omap' = omap[yoid \mapsto obj[f \mapsto xref]] \\
vmap' = vmap[x \mapsto yfref] \quad F' = (rid, bvar, omap', vmap') \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F'), H \rangle
\end{array}$$

SWAP-REGION-OBJECT

$$\begin{array}{c}
F = (rid, bvar, fomap, vmap, fid) \\
x \neq bvar \quad vmap(x) = \text{Old}(xoid) \quad \text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \\
\text{resolveFA}(y.f, \langle (S :: F), H \rangle) = yfref \quad \{\text{Old}(xoid), \text{Old}(yoid)\}.loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \\
H(rid) = (boid, romap, Open) \quad romap(yoid) = obj \quad obj' = obj[f \mapsto \text{Old}(xoid)] \\
R' = (boid, romap[yoid \mapsto obj'], Open) \quad F' = (rid, bvar, fomap, vmap[x \mapsto yfref]) \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F'), H[rid \mapsto R'] \rangle
\end{array}$$

SWAP-REGION-REGION

$$\begin{array}{c}
F = (rid, bvar, fomap, vmap, fid) \quad x \neq bvar \quad vmap(x) = \text{Rld}(xrid) \\
\text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \quad \text{resolveFA}(y.f, \langle (S :: F), H \rangle) = yfref \\
\text{Old}(yoid).loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \quad H(xrid) = (_, _, \text{Closed}) \\
H(rid) = (boid, romap, Open) \quad romap(yoid) = obj \quad obj' = obj[f \mapsto \text{Rld}(xrid)] \\
R' = (boid, romap[yoid \mapsto obj'], Open) \quad F' = (rid, bvar, fomap, vmap[x \mapsto yfref]) \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F'), H[rid \mapsto R'] \rangle
\end{array}$$

SWAP-REGION-BRIDGE

$$\begin{array}{c}
F = (rid, bvar, fomap, vmap, fid) \quad x = bvar \\
\text{resolveV}(y, \langle (S :: F), H \rangle) = \text{Old}(yoid) \quad \text{resolveFA}(y.f, \langle (S :: F), H \rangle) = \text{Old}(yfoid) \\
\{\text{Old}(yoid), \text{Old}(yfoid)\}.loc(\langle (S :: F), H \rangle) = \text{Rgn}(rid) \\
H(rid) = (boid, romap, Open) \quad romap(yoid) = obj \\
obj' = obj[f \mapsto \text{Old}(boid)] \quad R' = (yfoid, romap[yoid \mapsto obj'], Open) \\
\hline
\langle (S :: F), H \rangle \xrightarrow{x.f := y} \langle (S :: F), H[rid \mapsto R'] \rangle
\end{array}$$

ENTER

$$\frac{F = (rid, bvar, fomap, vmap, fid) \quad resolveFA(x.f, \langle S :: F, H \rangle) = Rld(rid') \quad H(rid') = (boid, romap, Closed) \quad F' = (rid', a, \emptyset, fid + 1)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{enter } x.f \ a} \langle \langle S :: F :: F' \rangle, H[rid' \mapsto (boid, romap, Open)] \rangle}$$

EXIT

$$\frac{F = (rid, _, _, _, _) \quad H(rid) = (boid, omap, Open)}{\langle \langle S :: F' :: F \rangle, H \rangle \xrightarrow{\text{exit}} \langle S :: F', H[rid \mapsto (boid, omap, Closed)] \rangle}$$

MAKE-OBJ-STACK

$$\frac{F = (rid, bvar, omap, vmap, fid) \quad \langle \langle S :: F \rangle, H \rangle.freshObjectId = oid}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{makeObjStack } x} \langle S :: (rid, bvar, omap[oid \mapsto \emptyset], vmap[x \mapsto Old(oid)], fid), H \rangle}$$

MAKE-OBJ-REGION

$$\frac{F = (rid, bvar, fomap, vmap, fid) \quad \langle \langle S :: F \rangle, H \rangle.freshObjectId = oid \quad H(rid) = (boid, romap, Open) \quad R = (boid, romap[oid \mapsto \emptyset], Open)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{makeObjRegion } x} \langle S :: (rid, bvar, fomap, vmap[x \mapsto Old(oid)], fid), H[rid \mapsto R] \rangle}$$

MAKE-REGION

$$\frac{F = (rid, bvar, omap, vmap, fid) \quad \langle \langle S :: F \rangle, H \rangle.freshObjectId = oid' \quad oid'.toRegionId = rid' \quad R' = (oid', [oid' \mapsto \emptyset], Closed) \quad F' = (rid, bvar, omap, vmap[x \mapsto Rld(rid')], fid)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{makeRegion } x} \langle S :: F', H[rid \mapsto R'] \rangle}$$

MERGE

$$\frac{F = (rid, bvar, omap, vmap, fid) \quad vmap(x) = Rld(rid') \quad H(rid) = (boid, romap, Open) \quad H(rid') = (boid', romap', Closed) \quad F' = (rid, bvar, omap, vmap[x \mapsto boid'], fid) \quad R = (boid, romap \cup romap', Open)}{\langle \langle S :: F \rangle, H \rangle \xrightarrow{\text{merge } x} \langle S :: F', H[rid' \mapsto \epsilon][rid \mapsto R] \rangle}$$

References